STAT 7900: Special Topic: Causal Analysis



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STAT 7900

Recap: Propensity Scores & Matching

1. Confounding

Treatment groups differ before treatment.

2. Propensity score

Summarize treatment assignment using baseline X.

3. Matching

Compare treated and control units with similar treatment probabilities.

Matching changes WHO we compare.

Recap: Propensity Scores & Matching

$$e(X) = P(T = 1 \mid X)$$

X

Pre-treatment covariates used to model treatment assignment.

$e(X)$

A single number between 0 and 1: estimated probability of treatment.

Goal

Create treated/control groups comparable on baseline X.

It is a treatment-assignment score, not an outcome-risk score.

What the Propensity Score is, and is Not

It IS

$$P(T = 1 \mid baseline\ X)$$

- A balancing score under the usual assumptions
- A tool for matching, weighting, stratification, adjustment

It is NOT

$$P(Y = 1 \mid X)$$

- Proof that exchangeability holds
- A magic cure for unmeasured confounding
- A reason to include post-treatment variables

Propensity methods can balance measured covariates, not variables you never measured.

Why Can One Number Help?

$$Y(t) \perp T \mid X \quad \Rightarrow \quad Y(t) \perp T \mid e(X)$$

Before

Matching directly on age, BP, diabetes, smoking, income, etc. quickly becomes difficult.

After

Use e(X) as a scalar summary of treatment assignment to find comparable units.

Important: the balancing result depends on the causal assumptions and a useful propensity-score model.

Which Covariates Belong in the PS Model?

Usually include

- True confounders
- Strong baseline predictors of outcome
- Pre-treatment variables related to treatment and/or outcome

Be cautious

- Weak instruments / treatment-only predictors can worsen variance
- Highly redundant predictors may not help balance

Do NOT include

- Post-treatment mediators
- Post-treatment colliders
- Variables caused by treatment

Use causal knowledge first; prediction accuracy is not the objective.

Estimating the Propensity Score in R

Logistic regression

```
ps_mod <- glm(T ~ age + bp + diabetes + smoking,
              data = dat, family = binomial())

dat$ps <- predict(ps_mod, type = "response")
```

Interpretation

A patient with ps = 0.78 had an estimated 78% probability of receiving treatment, given measured baseline X.

Model can be logistic regression, GAM, trees, boosting, etc., but balance remains the target.

Do not choose a PS model only because it has the highest AUC.

Poor Overlap: A Warning, Not Just a Nuisance

Example

Treated patient: $e(X) = 0.96$

Nearest control: $e(X) = 0.42$

These people did not have similar chances of receiving treatment.

What this means

- The data provide little information about the missing counterfactual.
- Matching may discard units.
- IPW will later create extreme weights.

Positivity is a causal assumption; overlap is its empirical footprint.

Matching: The Core Idea

Treated patient

Age 68

Diabetes: Yes

Baseline SBP: 158

$e(X) = 0.73$

Bad control

Age 35

Diabetes: No

Baseline SBP: 122

$e(X) = 0.14$

Better control

Age 66

Diabetes: Yes

Baseline SBP: 154

$e(X) = 0.70$

Match people who had similar treatment probabilities before treatment.

Nearest-Neighbor Matching

For treated i, choose control j minimizing $| e(X_i) - e(X_j) |$

Treated

$$e(X) = 0.71$$

Control A

$$e(X) = 0.69$$

Control B

$$e(X) = 0.43$$

Nearest neighbor would prefer Control A: but 'nearest' does not guarantee 'close enough'.

Calipers: Prevent Terrible Matches

Accept match only if $|e(X_i) - e(X_j)| < \text{caliper}$

Without a caliper

A treated unit may be matched to a distant control simply because that is the nearest available one.

With a caliper

Poor matches are rejected. Some treated units may remain unmatched.

Trade-off: better comparability can reduce the matched sample size.

Matching Choices Change the Analysis

$1:1$ *vs* $1:k$

$1:1$ is simple.

Using multiple controls can improve precision if good controls exist.

With replacement

A strong control can be reused.

Often improves match quality.

Requires correct variance handling.

Without replacement

Each control used once.

More intuitive sample, but later matches can become worse.

Matching is a design decision, not a single automatic procedure.

Before Matching vs After Matching

Before

Treated mean age: 68

Control mean age: 52

Diabetes: 42% vs 17%

Baseline SBP: 156 vs 139

Groups are not comparable.

After

Treated mean age: 67

Matched control: 66

Diabetes: 41% vs 39%

Baseline SBP: 155 vs 154

Much better baseline balance.

Matching is successful only if it improves balance on the baseline covariates.

What Estimand Does Matching Target?

Typical treated-to-control matching

Start with treated units.

Find comparable controls.

Estimate effect in the matched treated population.

Often naturally targets ATT.

But not always

Target depends on the matching design.

Trimming changes the population.

Full matching / weighting-like methods can target different estimands.

Always name the target population before interpreting the effect.

What Matching Does NOT Solve

Unmeasured confounding

Matching balances measured X.

Hidden common causes can still bias the effect.

Poor overlap

Some units may have no credible match.

Discarding them changes the target population.

Bad causal variable choice

Matching on mediators/colliders can create bias.

PS methods do not replace DAG reasoning.

Matching can make groups look similar.
Causal validity still requires assumptions.

Bridge: Matching → IPW → AIPW

Matching

Use $e(X)$ to FIND comparable units.

Some units may be discarded.

Often ATT-like.

IPW

Use $e(X)$ to REWEIGHT units.

Keep units but change how much each counts.

Can target ATE/ATT with different weights.

AIPW

Combine weighting with an outcome model.

Two modeling components.

Doubly robust under standard conditions.

Same propensity score idea: three different ways to build the causal comparison.

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IPW: changing each person’s contribution

Matching selects comparison participants.

Weighting assigns a numerical contribution to each observed participant.

The goal is to align covariate distributions for a chosen causal target.

We continue with training, baseline X, and 1978 earnings.

IPW uses treatment probabilities to construct the comparison.

Why inverse probabilities?

A treated person with $e(X) = 0.20$ represents a relatively rare assignment.

Giving that person weight $\frac{1}{0.20} = 5$ compensates for the selection probability.

A treated person with $e(X) = 0.80$ receives weight 1.25.

The same reasoning applies to controls using $1 - e(X)$.

Less likely observed assignments receive larger ATE weights.

ATE weights

Observed treatment	Probability of that assignment	ATE weight
$T = 1$	$e(X)$	$1/e(X)$
$T = 0$	$1 - e(X)$	$1/\{1 - e(X)\}$

$$w_i = \frac{T_i}{\hat{e}_i} + \frac{1 - T_i}{1 - \hat{e}_i}$$

Actual weights for job-training participants

Participant	T	Estimated score	ATE weight
NSW1	1	0.639	1.57
NSW11	1	0.689	1.45
PSID1	0	0.026	1.03
PSID16	0	0.056	1.06

For an untreated participant, use 1/(1 − score), not 1/score.

Why weighting balances the population

The observed treated covariate density is proportional to $e(x)f(x)$.

Multiplying by $1/e(x)$ leaves a density proportional to $f(x)$.

The control density is proportional to $\{1 - e(x)\}f(x)$.

Multiplying by $1/\{1 - e(x)\}$ also leaves $f(x)$.

With the true probabilities, both ATE-weighted arms target the same $f(x)$.

Why IPW identifies a treatment mean

$$E[TY/e(X) \mid X] = E[Y \mid T = 1, X].$$

Because $E[TY \mid X] = e(X)E[Y \mid T = 1, X]$.

Averaging over X gives $E[TY/e(X)] = E[Y(1)]$.

The control expression identifies $E[Y(0)]$ in the same way.

Exchangeability, consistency, and positivity justify the causal equality.

Weighted mean estimators

Horvitz-Thompson: $\hat{\mu}_1 = n^{-1} \Sigma T_i Y_i / \hat{e}_i$.

Normalized mean: $\hat{\mu}_1 = \Sigma T_i Y_i / \hat{e}_i \div \Sigma T_i / \hat{e}_i$. (Hajek)

Compute the corresponding control mean using $1 - \hat{e}_i$.

The effect estimate is $\hat{\tau} = \hat{\mu}_1 - \hat{\mu}_2$.

Our real-data calculations use normalized weighted means.

R: estimate the assignment model

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
ps_fit <- glm(treat ~ age + educ + race + married +
              nodegree + re74 + re75,
              family = binomial(), data = d)
d$ps <- predict(ps_fit, type = "response")
stopifnot(all(d$ps > 0 & d$ps < 1))
```

This check prevents invalid denominators. It does not verify population positivity.

R: ATE weights and point estimate

```
d$w_ate <- ifelse(d$treat == 1,
                  1 / d$ps, 1 / (1 - d$ps))
mu1 <- with(d, weighted.mean(re78[treat == 1],
                             w_ate[treat == 1]))
mu0 <- with(d, weighted.mean(re78[treat == 0],
                             w_ate[treat == 0]))
ate_ipw <- mu1 - mu0
c(mu1 = mu1, mu0 = mu0, ATE = ate_ipw)
```

The target is the overall population represented by these study data.

ATE weighting: real-data results

Quantity	Estimate
Weighted training mean	$6,648
Weighted comparison mean	$6,423
Weighted difference	About +$225

This is an estimate under the proposed model and assumptions, not a validated causal conclusion.

ATT weighting

Observed treatment	ATT weight	Purpose
$T = 1$	1	Retain the treated population
$T = 0$	$e(X)/\{1 - e(X)\}$	Make controls resemble treated units

ATT weights target the treated covariate distribution, not the full-population ATE.

R: ATT weights and point estimate

```
d$w_att <- ifelse(d$treat == 1,
                  1, d$ps / (1 - d$ps))
mu1_att <- with(d, weighted.mean(re78[treat == 1],
                                 w_att[treat == 1]))
mu0_att <- with(d, weighted.mean(re78[treat == 0],
                                 w_att[treat == 0]))
att_ipw <- mu1_att - mu0_att
c(mu1 = mu1_att, mu0 = mu0_att, ATT = att_ipw)
```

Here the treated mean stays $6,349. The weighted control mean is about $5,135.

ATE and ATT answer different questions

Target	Weighting estimate	Whose effect?
ATE	About +$225	Overall study population
ATT	About +$1,214	Treated participants

Different targets can produce different effects, even with the same data and propensity model.

Stabilized ATE weights

Treated: $P(T = 1)/e(X)$.

Control: $P(T = 0)/\{1 - e(X)\}$.

Use the observed treatment fraction to estimate the numerators.

The numerators multiply all weights within an arm by the same constant.

Normalized arm means are unchanged. Stabilization does not fix poor overlap.

Extreme weights and influence

For a treated person with $\hat{e} = 0.025$, the ATE weight is about 40.

In this real dataset, the maximum treated ATE weight is 40.08.

A small number of low-score treated people can drive the weighted treated mean.

Check the weight distribution separately in each arm.

Retaining everyone does not mean everyone contributes equally.

Effective sample size

$$ESS = (\Sigma w_i)^2 / \Sigma w_i^2.$$

Equal weights give ESS equal to the observed sample size.

A few dominant weights lower ESS.

Compute ESS within each treatment arm.

ESS describes weight concentration, not a new count of independent participants.

Weight diagnostics in the real data

Design	Treated n	Treated ESS	Control n	Control ESS
ATE weights	185	58.3	429	329.0
ATT weights	185	185.0	429	99.8

ATE weights concentrate influence among treated units. ATT weights concentrate it among controls.

R: inspect weights and ESS

```
ess <- function(w) sum(w)^2 / sum(w^2)
q <- c(0, .5, .9, .95, .99, 1)
tapply(d$w_ate, d$treat, quantile, probs = q)
tapply(d$w_ate, d$treat, ess)
tapply(d$w_att, d$treat, ess)
# Largest ATE weights and their underlying scores
ix <- order(d$w_ate, decreasing = TRUE)[1:5]
d[ix, c("treat", "ps", "w_ate", "age", "re74", "re75")]
```

Inspect the people behind the largest contributions.

R: calculate the chart's balance values

```
X <- data.frame(age=d$age, educ=d$educ, married=d$married,
 nodegree=d$nodegree, re74=d$re74, re75=d$re75,
 black=as.numeric(d$race == "black"))
A <- d$treat
sref <- sapply(X[A == 1, ], sd)
smd <- function(w) sapply(seq_along(X), function(j) {
 x <- X[[j]]
 abs(weighted.mean(x[A == 1], w[A == 1]) -
     weighted.mean(x[A == 0], w[A == 0])) / sref[j]
})
```

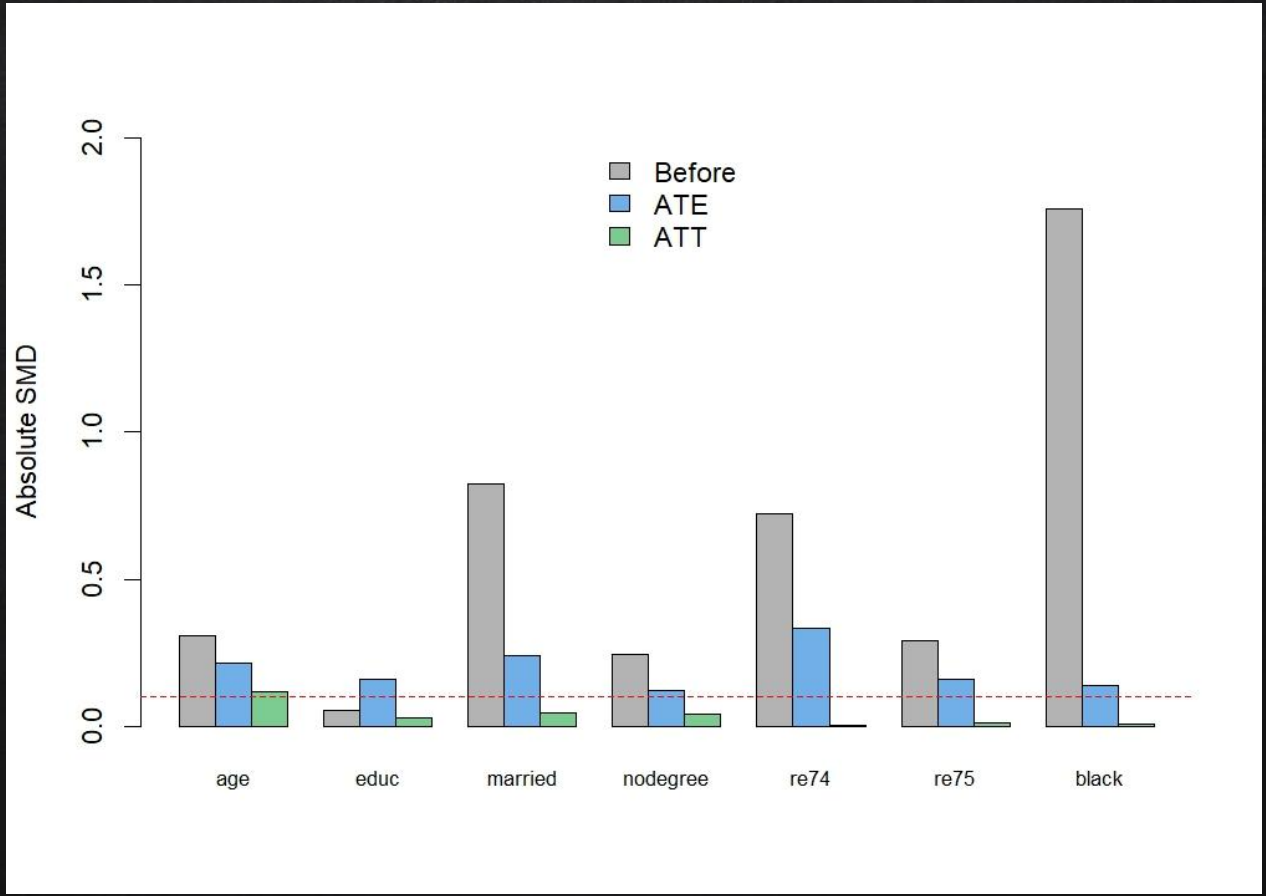
Use one fixed denominator before and after weighting.

R: reproduce the balance plot

```
balance <- rbind(
 Before = smd(rep(1, nrow(d))),
 ATE = smd(d$w_ate), ATT = smd(d$w_att))
barplot(balance, beside = TRUE,
 names.arg = names(X), cex.names = .75,
 col = c("grey70", "#71AFE8", "#7CCB91"),
 ylab = "Absolute SMD", ylim = c(0, 2),
 legend.text = rownames(balance),
 args.legend = list(x = "top", bty = "n"))
abline(h = .10, lty = 2, col = "red")
```

The dashed 0.10 line is a guideline, not a test of exchangeability.

Balance after weighting



Poor overlap: possible responses

Reconsider whether the full-population ATE is supported.

Improve the propensity specification if important modeled terms are missing.

Restriction changes the target population. Weight capping can introduce bias.

Recheck balance and sensitivity after any change.

A numerical weight modification cannot restore missing support.

Uncertainty and the next estimator

Use inference that accounts for the weighting analysis and any clustering.

A suitable refitting bootstrap can include propensity-model estimation for smooth IPW.

Sampling intervals do not quantify unmeasured confounding.

Next, AIPW combines outcome predictions with weighted residual corrections.

IPW depends on the treatment model. AIPW adds an outcome-model safeguard.

AIPW: why combine the two approaches?

G-computation predicts outcomes under training and no training.

IPW uses treatment probabilities to reconstruct comparable populations.

Each approach depends on a different statistical model.

AIPW combines predictions with weighted observed prediction errors.

Our target in this section is the same full-study-population ATE.

The two models

Model	Definition	In the job-training data
Outcome	$m_t(x) = E[Y \mid T = t, X = x]$	Predict 1978 earnings
Treatment	$e(x) = P(T = 1 \mid X = x)$	Predict training membership

Both models use the baseline adjustment set justified by the causal question.

Start with outcome predictions

For each person, predict $\widehat{m}_1(X_i)$ under training.

Also predict $\widehat{m}_0(X_\mathrm{i})$ under no training.

G-computation averages these predictions over all participants.

If the outcome model is wrong, prediction errors may systematically affect the mean.

AIPW starts with the same predictions, then adds a correction.

Observed residuals supply the correction

For a treated person, the observed training residual is $Y_i - \hat{m}_1(X_i)$.

For a control, the observed no-training residual is $Y_i - \hat{m}_0(X_i)$.

We only observe the residual in the arm received.

Inverse probabilities adjust which people supply those residuals.

The correction uses observed outcomes without inventing missing residuals.

The augmented training mean

$$\psi_{1i} = \hat{m}_1(X_i) + T_i\{Y_i - \hat{m}_1(X_i)\}/\hat{e}_i.$$

Every person contributes a predicted training outcome.

Only treated people contribute a training residual correction.

$$\hat{\mu}_{1,AIPW} = n^{-1}\Sigma_i \psi_{1i}.$$

Prediction + inverse-probability-weighted residual

The augmented no-training mean

$$\psi_{1i} = \hat{m}_0(X_i) + (1 - T_i)\{Y_i - \hat{m}_0(X_i)\}/(1 - \hat{e}_i).$$

Every person contributes a predicted no-training outcome.

Only controls contribute a no-training residual correction.

$$\hat{\mu}_{0,AIPW} = n^{-1}\Sigma_i \psi_{0i}.$$

Both means average over the same target population.

AIPW ATE

$$\hat{\tau}_{AIPW} = n^{-1}\Sigma_i [\psi_{1i} - \psi_{0i}].$$

Prediction contrast: $\hat{m}_1(X_i) - \hat{m}_0(X_i)$.

Add the treated residual correction $T_i\{Y_i - \hat{m}_1(X_i)\}/\hat{e}_i$.

Subtract the control residual correction $(1 - T_i)\{Y_i - \hat{m}_0(X_i)\}/(1 - \hat{e}_i)$.

The control correction enters with a minus sign.

Worked real participant: NSW1

Quantity	Value
Treatment and observed earnings	$T = 1, Y = \$9{,}930$
Predicted training earnings	$\$5{,}964$
Estimated treatment probability	0.639
Augmented training contribution	$5964 + (9930 - 5964)/0.639 \approx \$12{,}173$

Its control contribution is the prediction $\hat{m}_0(X) \approx \$4,416$.

Worked real participant: PSID1

Quantity	Value
Treatment and observed earnings	T=0, Y=$25,565
Predicted no-training earnings	$15,935
Estimated treatment probability	0.026
Augmented control contribution	15935 + (25565 − 15935)/(1 − 0.026) ≈ $25,823

Its training contribution is the prediction $\hat{m}_1(X) \approx \$17,483.$

Why it works: a correct outcome model

Suppose $\widehat{m}_1(X)$ converges to the true $m_1(X)$.

Then $E[Y - m_1(X) \mid T = 1, X] = 0$.

The weighted residual correction has conditional expectation zero.

The average prediction identifies $E[Y(1)]$.

A correct outcome model makes the correction mean-zero.

Why it works: a correct propensity model

Suppose $\hat{e}(X)$ converges to the true $e(X)$.

Let $\tilde{m}_1(X)$ be a possibly wrong outcome-model limit.

$$E[\tilde{m}_1(X) + T\{Y - \tilde{m}_1(X)\}/e(X) \mid X]$$

$$= \tilde{m}_1(X) + m_1(X) - \tilde{m}_1(X) = m_1(X).$$

Correct assignment probabilities remove the limiting prediction error.

Two errors interact in the remaining bias

Let g(X) be the propensity-model limit.

Let $\tilde{m}_1(X)$ be the outcome-model limit.

Bias of the augmented training mean:

$$E\left[\frac{\{g(X) - e(X)\}\{\tilde{m}_1(X) - m_1(X)\}}{g(X)}\right].$$

If either error is zero, this bias term vanishes.

What double robustness does not fix

An unmeasured confounder can invalidate both model-based causal interpretations.

Missing support can prevent identification for part of the target population.

Scores near 0 or 1 can create extreme residual corrections.

Two misspecified models can still yield a biased estimate.

Double robustness concerns statistical models, not a replacement for causal assumptions.

R: fit the outcome and treatment models

```
library(MatchIt)
data("lalonde", package = "MatchIt")
d <- lalonde
f_x <- ~ age + educ + race + married + nodegree + re74 + re75
fit_y <- lm(update(f_x, re78 ~ treat + .), data = d)
fit_e <- glm(update(f_x, treat ~ .),
             family = binomial(), data = d)
m1 <- predict(fit_y, transform(d, treat = 1))
m0 <- predict(fit_y, transform(d, treat = 0))
e <- predict(fit_e, type = "response")
```

The outcome model predicts earnings. The treatment model predicts assignment.

R: calculate augmented contributions

```
A <- d$treat
Y <- d$re78
stopifnot(all(e > 0 & e < 1))
psi1 <- m1 + A * (Y - m1) / e
psi0 <- m0 + (1 - A) * (Y - m0) / (1 - e)
psi <- psi1 - psi0
ate_aipw <- mean(psi)
c(mu1 = mean(psi1), mu0 = mean(psi0), ATE = ate_aipw)
```

Prediction errors use observed minus predicted outcomes.

R: compare the same ATE target

```
ate_g <- mean(m1 - m0)
w <- ifelse(A == 1, 1 / e, 1 / (1 - e))
ate_ipw <- weighted.mean(Y[A == 1], w[A == 1]) -
           weighted.mean(Y[A == 0], w[A == 0])
c(G_computation = ate_g,
  IPW_normalized = ate_ipw,
  AIPW = ate_aipw)
```

All three calculations target the overall study-population ATE.

Real-data estimates

Method	ATE estimate	Main statistical dependence
G-computation	About +$1,548	Outcome model
Normalized IPW	About +$225	Propensity model
AIPW	About +$887	Either model for consistency

AIPW's position between these estimates does not prove it is closest to the truth.

What changed relative to g-computation?

Average predicted training earnings $\approx \$7{,}875$.

Training residual correction $\approx -\$557$, giving $\$7{,}318$.

Average predicted control earnings $\approx \$6{,}326$.

Control residual correction $\approx +\$104$, giving $\$6{,}431$.

AIPW difference: $7318 - 6431 \approx \$887$.

Uncertainty: what the simple score SE requires

When both nuisance functions are consistently estimated and suitable regularity holds:

$\hat{\phi}_i = \psi_i - \hat{\tau}_{AIPW}$ approximates an influence contribution.

$SE \approx sd(\psi_i)/\sqrt{n}$, with a normal confidence interval.

Consistency with one correct model alone does not justify this SE universally.

Use a variance procedure justified for the fitted models and target.

R: a refitting bootstrap function

```
estimate_aipw <- function(z) {
 fy <- lm(re78 ~ treat + age + educ + race + married +
          nodegree + re74 + re75, data = z)
 fe <- glm(treat ~ age + educ + race + married +
           nodegree + re74 + re75, binomial(), data = z)
 p <- predict(fe, type = "response")
 y1 <- predict(fy, transform(z, treat = 1))
 y0 <- predict(fy, transform(z, treat = 0))
 A <- z$treat; Y <- z$re78
 mean(y1 - y0 + A*(Y-y1)/p - (1-A)*(Y-y0)/(1-p))
}
```

Each resample refits both models before calculating AIPW.

R: bootstrap uncertainty

```
set.seed(7900)
B <- 500
boot_est <- replicate(B, {
 idx <- sample.int(nrow(d), replace = TRUE)
 estimate_aipw(d[idx, ])
})
stopifnot(all(is.finite(boot_est)))
boot_se <- sd(boot_est)
ci <- ate_aipw + c(-1, 1) * qnorm(.975) * boot_se
c(ATE = ate_aipw, SE = boot_se, lower = ci[1], upper = ci[2])
```

Inspect resampling stability and convergence warnings before reporting the interval.

Flexible models and cross-fitting

Flexible models can estimate nonlinear outcome and assignment functions.

Fit nuisance models outside each held-out fold.

Compute each held-out participant's score using those excluded-data fits.

Average all held-out scores and assess the required convergence conditions.

Cross-fitting limits overfitting effects. It does not fix confounding or support.

AIPW: the main lesson

Outcome predictions supply the treatment and control means.

Weighted factual residuals correct systematic prediction errors.

Either correctly specified nuisance side can protect consistency.

Causal assumptions, overlap, and suitable uncertainty quantification remain essential.

The final report should state the target, models, diagnostics, estimate, and limitations.

Does the Treatment Effect Differ Across People?

$$\mathrm{ATE} = \mathrm{E}[\mathrm{Y}(1) - \mathrm{Y}(0)]$$

So far

Estimate one average causal effect for the target population.

Next

Ask whether the effect changes with baseline characteristics X.

This is heterogeneous treatment effect (HTE) analysis.

Why the Average Can Be Misleading

Subgroup A

Treatment effect = +10

Subgroup B

Treatment effect = −10

If both groups are equally common: $\mathrm{ATE} = 0$

An average effect of zero does not mean the treatment does nothing for everyone.

From ATE to CATE

$$CATE(x) = E[Y(1) - Y(0) \mid X = x]$$

ATE

How much does treatment help on average?

CATE

How much does treatment help for people with $X = x$?

Examples

Age, disease severity, biomarker, baseline risk.

ATE averages over X; CATE conditions on X.

CATE Is Not an Individual Treatment Effect

Individual treatment effect

$$Y_i(1) - Y_i(0)$$

Defined for one specific person.

Fundamentally unobservable without strong structural assumptions.

CATE(x)

Average treatment effect among people with $X = x$.

Still an average, just a more local one.

Usually a more realistic statistical target.

HTE typically means variation in average effects across covariate profiles.

Effect Modifier ≠ Confounder

Confounder

Affects treatment assignment and outcome.

Creates a noncausal backdoor path.

Needs control for identification.

Effect modifier

Changes the magnitude or direction of the treatment effect.

Answers: 'For whom does treatment work better?'

May or may not also be a confounder.

A variable can be both: but confounding and effect modification are different causal roles.

Real-World Heterogeneity: Medicine

Baseline risk

A statin may prevent more absolute events in high-risk patients.

Biomarker

A targeted cancer therapy may work mainly in biomarker-positive tumors.

Age / frailty

Benefit and toxicity may differ across age or frailty levels.

Clinical decisions are often about who benefits most.

Real-World Heterogeneity: Education and Policy

Education

Tutoring may benefit lower-performing students more.

Technology interventions may differ by prior achievement or access.

Policy

Job training may help workers with weak labor-market attachment more.

Cash transfers may have different effects by baseline income.

HTE turns one average policy effect into a targeting question.

A Simple Two-Subgroup Example

Group	Control mean	Treatment mean	Effect
Low baseline risk	20	18	-2
High baseline risk	40	30	-10

Same treatment, different causal effect across baseline risk.

Starting Point: Treatment × Covariate Interaction

$$Y = \beta_0 + \beta_1 T + \beta_2 X + \beta_3 (T \times X) + \epsilon$$

β_1

Treatment effect when $X = 0$.

β_2

Outcome association with X in the control arm.

β_3

How the treatment effect changes as X changes.

$$\tau(x) = \beta_1 + \beta_3 x$$

How to Read an Interaction

Suppose

$\beta_1 = -2$ and $\beta_3 = -1.5$

Then

At $X = 0$: effect = -2

At $X = 1$: effect = -3.5

At $X = 2$: effect = -5

A nonzero interaction means the treatment effect changes with X.

Binary Effect Modifier Example

No diabetes $(X = 0)$

Estimated treatment effect $=$ $-3\ mmHg$

Diabetes $(X = 1)$

Estimated treatment effect = $-\ 8\ mmHg$

Interaction asks whether -8 and -3 differ beyond sampling variation.

Subgroup Analysis Can Go Wrong

Small subgroups

Noisy estimates

Wide confidence intervals

Unstable signs

Post hoc fishing

Try enough subgroups and something will look significant.

False discoveries multiply.

Arbitrary cutpoints

Age 59 vs 60

Risk 9.9% vs 10.1%

Artificial discontinuities

HTE is harder than ATE because we estimate many local effects.

Prediction Is Not Treatment-Effect Heterogeneity

Prognostic model

Predicts Y well.

Finds who is high risk.

High risk does not necessarily mean larger treatment benefit.

HTE model

Targets $Y(1) - Y(0)$ as a function of X.

Finds who benefits more or less from treatment.

High outcome risk ≠ high treatment benefit.

S-Learner: One Flexible Outcome Model

Fit one model: $m(T, X) = E[Y \mid T, X]$

Step 1

Fit one regression / ML model using T and X.

Step 2

Predict each person twice: once with $T = 1$, once with $T = 0$.

Step 3

Subtract the two predictions.

$$\hat{\tau}(x) = \hat{m}(1, x) - \hat{m}(0, x)$$

T-Learner: Separate Models by Treatment Arm

$$m_1(x) = E[Y|T = 1, X = x] \qquad m_0(x) = E[Y|T = 0, X = x]$$

Treatment model

Fit outcome model using treated observations only.

Control model

Fit outcome model using control observations only.

CATE

$$\hat{\tau}(x) = \hat{m}(1, x) - \hat{m}(0, x)$$

T-learner allows the two response surfaces to be very different.

Causal Forests: A Preview

Ordinary random forest

Splits data to improve prediction of Y.

Causal forest

Splits data to find regions where treatment effects differ.

Output

Estimated CATE by covariate profile.

Variable importance for heterogeneity.

Uncertainty tools.

Goal: not 'who has high Y?' but 'where is the treatment contrast different?'

HTE Still Needs the Same Causal Assumptions

$$Y(t) \perp T \mid X \;+\; \text{positivity} \;+\; \text{consistency}$$

Exchangeability

Within X, treated and untreated are comparable for potential outcomes.

Positivity

Relevant X-profiles need treatment and control support.

Consistency

Observed outcome corresponds to the treatment actually received.

Flexible ML does not make unmeasured confounding disappear.

Next Class: HTE in Detail

1. Regression interactions

Binary and continuous effect modifiers

How to plot τ(x)

Confidence intervals

2. Meta-learners

S-learner

T-learner

X-learner

When each is useful

3. Modern HTE

Causal forests

Sample splitting / validation

Clinical targeting

R implementation

For today, remember: 'For whom is the treatment effect different?'